INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY, HYDERABAD

RAG over the EU AI Act

Baseline Retrieval vs. Cross-Encoder Re-Ranking

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RAG Assignment

CL3.410 | Language Models and Agents

October 2026

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1 Setup

Document. I used the EU AI Act (Regulation (EU) 2024/1689), the official 144-page PDF from the Official Journal. It suits this assignment because:

- it has prohibitions with narrow exceptions, e.g. emotion recognition at work is banned unless it is for medical or safety reasons;
- the rules change with the role: provider, deployer, importer and distributor each have their own duties;
- it has numbers that are easy to mix up, e.g. the fine for a start-up is capped in a different paragraph from the general fine;
- it starts with 180 *recitals*: plain-language explanations that are not binding but talk about the same topics as the articles, so they make natural distractors.

Cleaning. I extracted the text with PyMuPDF and removed the running header and footer on every page, as well as the footnotes, which use a smaller font. The recital numbers "(1)", "(2)", … use that same small font, so the footnote filter removed them too at first. I now keep a small "(n)" when it sits on the same line as body text. One superscript also had to be restored: the systemic-risk threshold of 10^{25} FLOPs came out as "1025".

Chunking. The chunks follow the structure of the law: one chunk per recital, articles split by numbered paragraph, and annexes split by numbered point. Paragraphs are packed together up to 230 tokens. A paragraph that is too long is split at its points (a), (b), …, and each piece repeats the paragraph's opening sentence. Every chunk starts with a header such as `[Article 99 – Penalties, para 5–6]`. I kept every chunk under 256 tokens, because that is all the bi-encoder reads. This gives 747 chunks: 289 recital, 387 article and 71 annex (Fig. 1).

Models.

- **Bi-encoder:** `all-MiniLM-L6-v2`, searched with an exact FAISS inner-product index (cosine similarity on normalised vectors).
- **Cross-encoder:** `ms-marco-MiniLM-L-6-v2`.
- **LLM:** `qwen2.5:7b` through Ollama, temperature 0, fixed seed.

The two encoders have the same 6-layer MiniLM backbone, so the difference between the pipelines comes from the architecture, not from model size.

Prompt. Both pipelines use the same system prompt. It tells the model to:

- answer only from the three excerpts;
- give the direct answer first, then any condition, exception, deadline or amount;
- cite the excerpt headers;
- reply "The provided context does not specify this" when the answer is not there.

My first version only said "cite the headers". The model then replied with nothing but a header in about one answer in eight, in every pipeline. I changed the wording for all pipelines and regenerated every answer.

Questions and labels. I wrote 49 questions the way a normal user would ask them, grouped as negation, exception, role limit, keyword trap, corner case, scope, penalty and control. Each question is labelled with the binding clause that answers it, matched by article, paragraph and a short anchor phrase. The anchor is needed because Article 5(1) alone spans six chunks. Q11 (general fine *and* start-up cap) and Q40 each need two clauses.

I report three retrieval metrics:

- **hit@3:** a gold clause is among the three chunks sent to the LLM;
- **all@3:** every required clause is among them;
- **MRR@10**.

For the answers, each question has 1–3 key facts (e.g. "10 days"). *Key-fact coverage* is the share of them found in the answer. It is only a string-match proxy, so I also read the answers myself. Appendix A lists every question with its rank in each pipeline.

2 Bi-encoder vs. cross-encoder

Bi-encoder. A bi-encoder encodes the question and each chunk *separately* into one vector each, by mean-pooling the token outputs. Relevance is the cosine similarity of the two vectors. All chunk vectors can be computed once and stored, so a query costs one forward pass plus a vector search. The price is that the question and the chunk never interact. A single token such as "not", "except" or "deployer" is averaged in with about 200 others and barely moves the vector.

Cross-encoder. A cross-encoder reads the pair *together*, as `[CLS] question [SEP] chunk`. Self-attention runs across both, so every question token can attend to every chunk token, and a head on top outputs one relevance score. This lets it notice that the question is about importers and this paragraph is about importers, or that a sentence is the exception rather than the rule. The cost is that nothing can be precomputed: every pair needs its own full forward pass.

How much they disagree. Fig. 2a plots both scores for all 490 question–candidate pairs. They agree on the broad order (Spearman $\rho = 0.72$) but often disagree inside the top 10. In Fig. 2b, 96 of the 147 bi-encoder top-3 chunks (about two thirds) stay in the top 3 after re-ranking. The rest are swapped with chunks from ranks 4–10.

3 Rank shifts and answers

On retrieval, re-ranking helped only a little on average (Table 1). The answers improved more: Answer 2 had more key facts than Answer 1 for 7 questions, fewer for 1, and

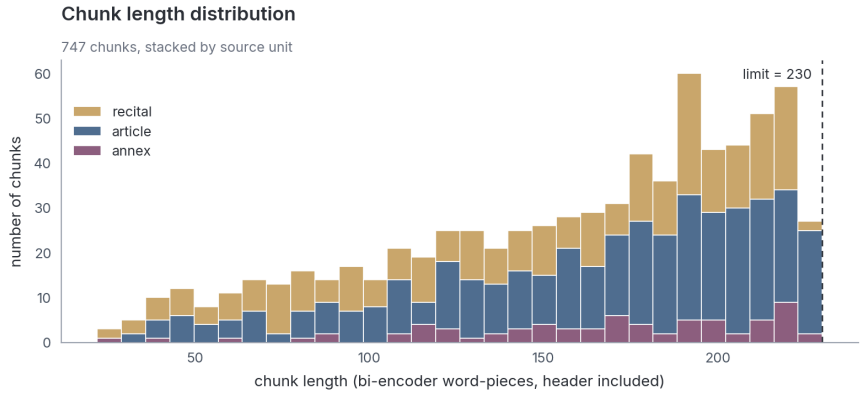
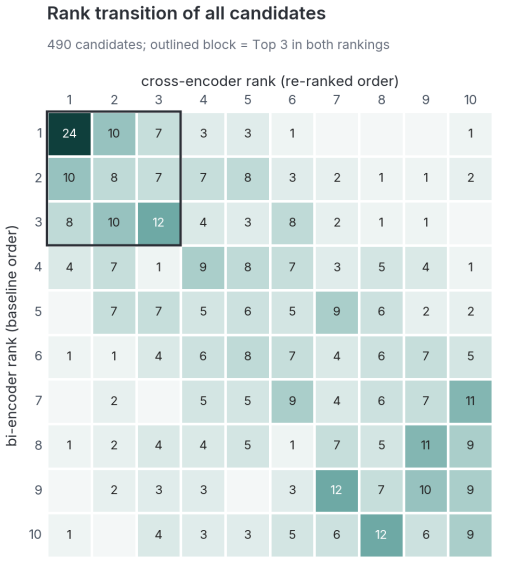


Figure 1. Chunk lengths in bi-encoder tokens, header included. The dashed line marks the 230-token packing limit.



(a) Bi-encoder cosine vs. cross-encoder logit; gold chunks in teal.



(b) Bi-encoder rank (rows) vs. rank after re-ranking (columns).

Figure 2. The two models over the same 490 candidates.

49 questions	P1: bi-encoder	P2: re-ranked
hit@3	0.633	0.653
all@3	0.612	0.633
MRR@10	0.474	0.502
Key-fact coverage	0.673	0.769

Table 1. Pipeline 1 vs. Pipeline 2 on all questions.

the same for 41 (Fig. 6). Fig. 3 shows the gold rank of every question before and after re-ranking.

3.1 Failures that re-ranking fixed

Four questions had their gold clause at rank 4–10 under the bi-encoder and in the top 3 after re-ranking (Table 2).

Q16 (role limit). All the paragraphs of Article 23 repeat "importer" and "high-risk AI system", so their vectors look almost the same. The bi-encoder ranked paragraphs 2–4 above paragraph 1, even though paragraph 1 is the one that lists the checks. With paragraph 2 on top, Answer 1 got the rule backwards:

> **Answer 1** *"An importer must ensure that a high-risk AI system is not in conformity with the Regulation or is falsified before placing it on the market."*

Answer 2 listed the real checks: the conformity assessment, the technical documentation, the CE marking and the declaration of conformity.

Q02 (keyword trap, negation). The question says "face recognition database" and "scraping photos". The law says "untargeted scraping of facial images". The bi-encoder's top 3 were Recital 43, Recital 30 and the *biometric categorisation* part of Article 5, which shares the vocabulary but is about something else. After re-ranking, the actual ban in Article 5(1)(e) was second. Both answers said "not legal", but Answer 1 could only lean on a recital that says the practice "should be prohibited".

Q27 (negation with an exception). Only the re-ranked context contained Article 5(1)(g), so only Answer 2 could state the exception correctly:

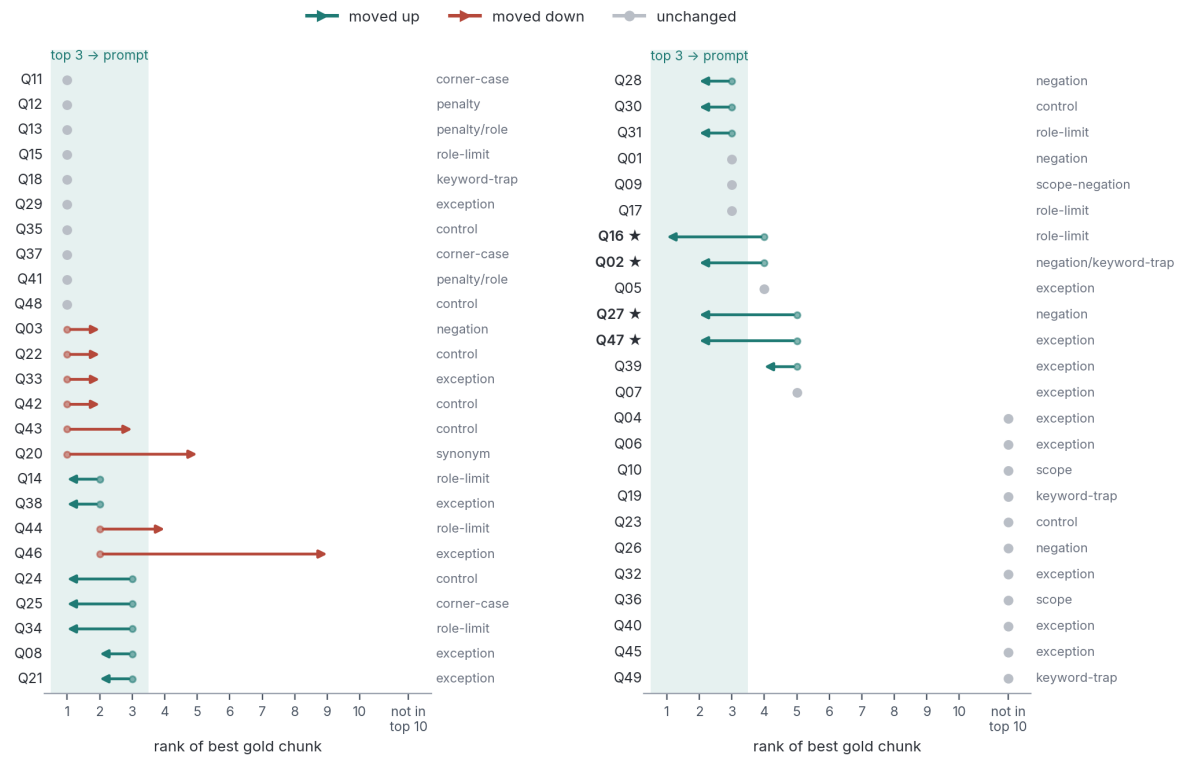


Figure 3. Gold-clause rank for every question, with an arrow from the bi-encoder rank to the re-ranked rank. The shaded band is the top 3 that reaches the prompt. ⋆ marks clauses promoted from rank 4–10 into the top 3.

	Type	Question	Gold clause	Rank
Q16	role limit	What must an importer check before placing a high-risk AI system on the market?	Art 23(1)	#4 → #1
Q02	negation / keyword trap	Is it legal to build a face recognition database by scraping photos from the internet?	Art 5(1)(e)	#4 → #2
Q27	negation + exception	Can a company use biometric data to sort people by their race or sexual orientation?	Art 5(1)(g)	#5 → #2
Q47	exception	Are AI systems used for detecting financial fraud classified as high-risk?	Annex III 5(b)	#5 → #2

Table 2. Failures that re-ranking fixed: the gold clause was buried at rank 4–5 and promoted into the top 3.

> **Answer 2** *"…the prohibition does not cover lawful labelling or filtering of biometric datasets acquired in line with Union or national law."*

Answer 1 had only Recital 30, and it turned the exception into something looser:

> **Answer 1** *"…sorting individuals according to less sensitive characteristics like hair colour or eye colour for law enforcement purposes is allowed."*

Q47 (exception inside a list). The rule is one clause inside Annex III 5(b): credit scoring is high-risk "with the exception of AI systems used for the purpose of detecting financial fraud". The bi-encoder's top 3 were two recitals and the wrong paragraph of Article 6. Both final answers say "not high-risk", because one recital happens to say something similar, but only the re-ranked context had the binding exception.

3.2 Where re-ranking made things worse

Q20. "Do people have to be told that they are chatting with a bot?" The bi-encoder had Article 50(1) at rank 1. The cross-encoder moved it to rank 5 and put three recitals on top. The answer was still roughly right, but it could only cite Recital 132. "Chatting with a bot" shares almost no words with "AI systems intended to interact directly with natural persons". My guess is that the cross-encoder, trained on web search, preferred recitals written in everyday language.

Q46. "Can police use biometric identification to find a missing child?" The exception list of Article 5(1)(h) fell #2 → #9. Answer 1 quoted the "missing persons" exception. Answer 2 had only Recital 33 and called it a rule about *post-remote* identification, which is a different kind of system.

Q23: a role-limit failure caused by re-ranking. "How long must providers keep the technical documentation?" The cross-encoder put the *importer's* 10-year rule (Article 23) at rank 1, and Answer 2 then said:

> **Answer 2** *"Providers shall keep the technical documentation … for a period of 10 years …, as per [Article 23 – Obligations of importers]. This obligation is similar for providers and importers."*

The number is right by luck. The source belongs to the wrong role, and the last sentence is made up.

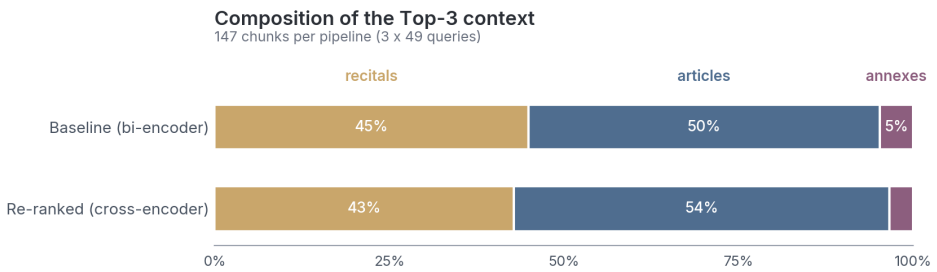


Figure 4. Share of recitals, articles and annexes among the 147 chunks placed in the prompt (3 per question).

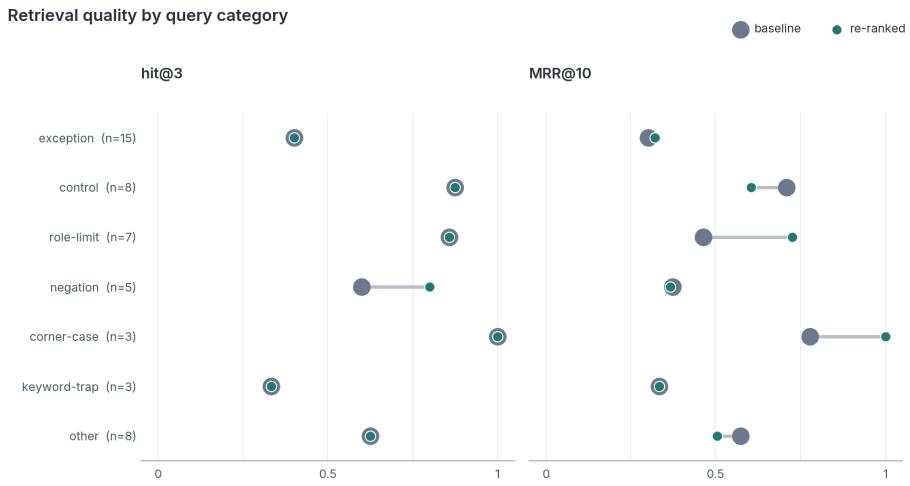


Figure 5. hit@3 and MRR@10 per question type, before and after re-ranking. Types with fewer than 3 questions are pooled as "other".

3.3 When the first stage misses

For 11 of the 49 questions the gold clause was never in the bi-encoder top 10, so the cross-encoder never saw it. The clearest case is Q11, "What is the maximum fine for a start-up that puts a prohibited AI system on the market?" The top 10 had Article 99(3): €35M or 7%, "whichever is higher". It did not have Article 99(6): for SMEs and start-ups the fine is "whichever is lower". Answer 1 abstained, and Answer 2 stated the general rule, which is wrong for a start-up. The misstatement is caused only by a missing chunk.

3.4 The recitals trap both models

The rank-1 chunk was a recital for 25 of the 49 questions under the bi-encoder, and for 27 after re-ranking. Over the whole top 3, recitals make up 45% of the context before re-ranking and 43% after (Fig. 4). So the cross-encoder does not prefer recitals in general. Recitals look like good answers to *both* models because they are written as explanations, but they often leave out the exact condition, amount or exception. Section 6 goes after this.

By question type (Fig. 5), re-ranking helped most for role limits (MRR $0.46 \rightarrow 0.73$) and negation (hit@3 $0.6 \rightarrow 0.8$). Exceptions stayed the weakest group with either model, because the exception clause was often not in the top 10 at all.

Stage	median	p95	mean
Query embedding	4.53	5.48	4.70
FAISS search (top 10)	0.10	0.13	0.11
Cross-encoder, 10 pairs, GPU	17.62	19.77	17.78
Cross-encoder, 10 pairs, CPU	142.08	179.29	145.47
P1 retrieval	**4.64**	**5.58**	**4.81**
P2 retrieval, GPU	**22.30**	**25.04**	**22.59**
P2 retrieval, CPU	146.63	184.69	150.27
LLM answer, P1	5416.29	9460.26	5546.89
LLM answer, P2	5197.95	9531.60	5639.54
End to end, P1	5421.40	9464.91	5551.70
End to end, P2	5220.68	9554.23	5662.13

Table 3. Latency per stage in milliseconds. P1 retrieval = query embedding + FAISS search; P2 adds the cross-encoder over 10 pairs.

4 Latency

All timings are per query on an RTX 4060 laptop GPU: 5 warm-up runs, then each question timed 3 times (147 runs). GPU stages are wrapped in `torch.cuda.synchronize()` so that queued work is counted.

- Re-ranking makes retrieval about 5× slower on the GPU and about 32× slower when the cross-encoder runs on the CPU (Table 3).
- Almost all of the extra time is the cross-encoder's 10 full transformer passes.
- End to end it hardly matters: the LLM takes about 5 s, so the extra ∼17 ms is well under 1% of what the user waits.

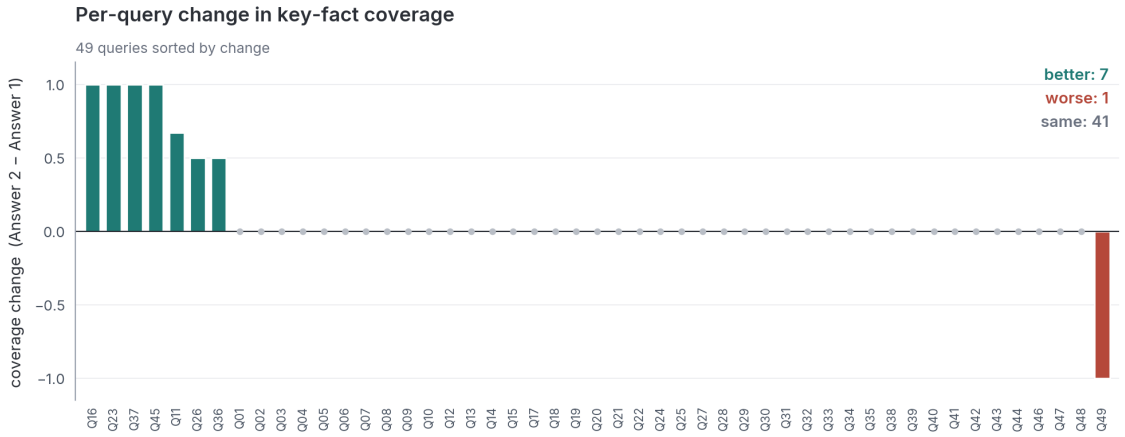


Figure 6. Change in key-fact coverage per question (Answer 2 minus Answer 1). The one drop, Q49, is a date question where both contexts missed Article 113.

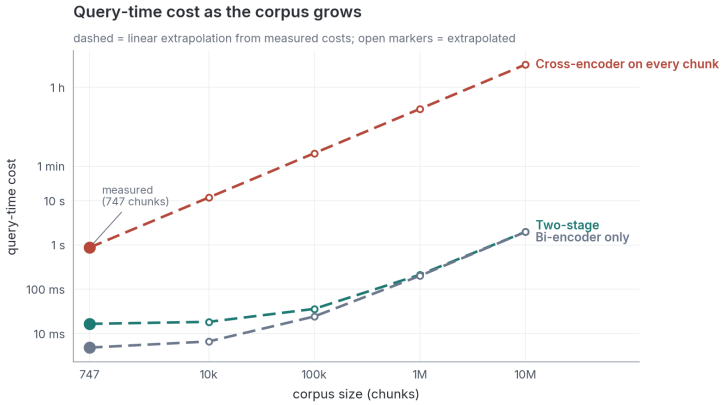


Figure 7. Query-time cost as the collection grows (log–log). Filled marker = measured, open = extrapolated.

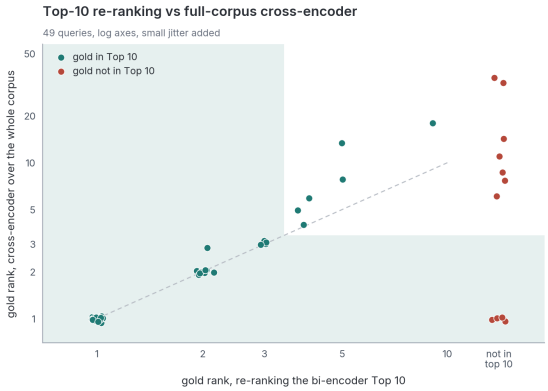


Figure 8. Gold rank after re-ranking the top 10 (x) vs. cross-encoding all 747 chunks (y). Red: not in the bi-encoder top 10.

Chunks	Bi-enc.	CE on all	Two-stage	CE/bi
747	4.8 ms	880.8 ms	16.6 ms	182×
10,000	6.7 ms	11.8 s	18.5 ms	1,769×
100,000	24.4 ms	117.9 s	36.2 ms	4,835×
1,000,000	201.6 ms	19.7 min	213.4 ms	5,849×
10,000,000	2.0 s	3.3 h	2.0 s	5,974×

Table 4. Query-time cost, measured at 747 chunks and extrapolated linearly (flat index).

Fig. 9 shows the medians and the spread of every run.

5 Why not cross-encode everything?

The cross-encoder's cost grows with the size of the collection on *every* query, and none of the work can be reused. I timed it over all 747 chunks for one query: about 881 ms, or 1.18 ms per chunk. The whole bi-encoder retrieval for the same query took about 4.8 ms. So even for one document, scoring everything is about 180× slower. Table 4 and Fig. 7 extrapolate this linearly.

The reason is what can be stored ahead of time:

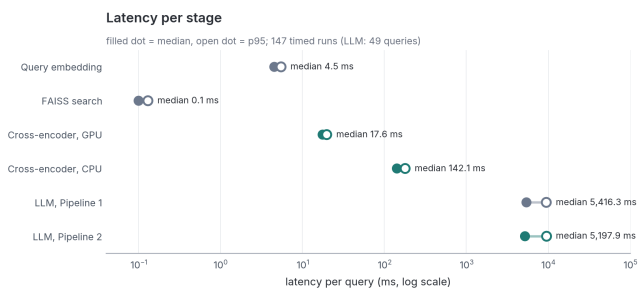
- **Bi-encoder:** the document vectors are computed once, offline. A query costs one embedding plus a search, and an approximate index such as HNSW makes that search grow much more slowly than the collection.
- **Cross-encoder:** the score depends on the question and the chunk together, so nothing can be computed before the question arrives. A new question means N new forward passes.

Sentence-BERT gives the classic example: finding the closest pair among 10,000 sentences takes about 65 hours with a BERT cross-encoder and about 5 seconds with SBERT. So retrieval is done in two stages: a cheap model narrows the collection to a few dozen candidates, and the expensive model only orders those.

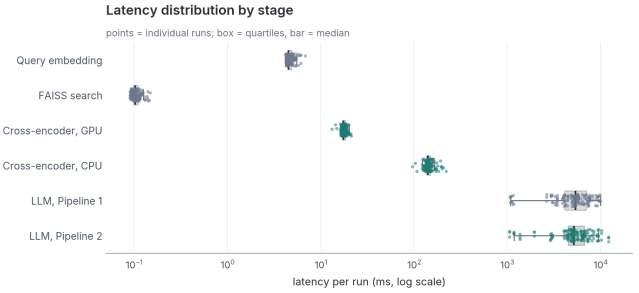
The two-stage design has a catch, and my results show it. When I let the cross-encoder rank the whole corpus, it put several clauses that the bi-encoder had missed completely at rank 1: Q23 (rank 41 under the bi-encoder), Q06 (16) and Q26 (14). These are shown in Fig. 8. So the cross-encoder can only be as good as the candidates the first stage hands it, which is why I ran the experiments in the next section.

6 Everything else I tried

The assignment fixes Pipelines 1 and 2, and everything above uses them exactly as specified. On top of that, I changed one design choice at a time and measured retrieval



(a) Median (filled) and p95 (open).



(b) Every timed run.

Figure 9. Latency per stage, log scale.

only, without the LLM. To avoid tuning on my own test questions, I chose configurations on Q01–Q24 (dev) and kept Q25–Q49 (test) only for reporting. Fig. 10 lists every configuration I ran, and Fig. 11 compares them.

Chunk size. Smaller chunks ($\leq$120 tokens) helped a little after re-ranking (MRR 0.558 vs. 0.502). Bigger chunks ($\leq$400) were clearly worse (0.444), because MiniLM silently drops everything after 256 tokens, and 216 of the 467 chunks were cut.

Chunk header. Removing the `[Article ...]` line lowered first-stage hit@3 from 0.633 to 0.469. The article title carries a lot of the meaning.

Binding text only. Dropping the recitals gave the biggest single gain (MRR 0.736). One caveat: my gold labels always point to binding text, so part of this gain is built in. That is why I also compared the *answers* below.

Candidate pool. Re-ranking the top 20, 30 or 50 instead of the top 10 raised the recall of the pool from 0.776 to 0.898, and MRR to 0.552.

First stage. bge-base and e5-base helped about as much as a bigger pool (MRR $\approx$ 0.56), while bge-small was no better than MiniLM. Merging dense results with BM25 by reciprocal rank fusion gave the best first-stage hit@3 among these first-stage changes (0.694, against 0.633). BM25 catches exact words such as "start-ups", "10 years" or "polygraphs".

Bigger re-rankers. MiniLM-L12, bge-reranker-base and bge-reranker-v2-m3 helped only a little *on their own* (MRR 0.54–0.57), because they still saw the same 10 candidates.

Combinations. Binding text + hybrid + top 30 brought the recall of the pool to 0.98. With MiniLM-L12 on top, MRR reached 0.818, and with bge-reranker-v2-m3 it reached 0.828, with almost the same score on dev and test.

6.1 Pipeline 3

bge-reranker-v2-m3 had the best dev score, but it has 568M parameters and does not fit on my 8 GB GPU next to the LLM. MiniLM-L12 was almost as good, so Pipeline 3 is:

1. binding text only;
2. dense + BM25 merged with reciprocal rank fusion;
3. top 30 candidates;
4. re-ranked with `ms-marco-MiniLM-L12-v2`;
5. top 3 into the *same* prompt.

	hit@3	all@3	MRR@10	Cover.	ms
P1	0.633	0.612	0.474	0.673	4.6
P2	0.653	0.633	0.502	0.769	22.3
P3	0.837	0.796	0.818	0.810	88.1

Table 5. All three pipelines on the 49 questions. Cover. = key-fact coverage of the answer; ms = median retrieval time. P3's time comes from single runs and uses an unoptimised Python BM25.

On the held-out questions, MRR@10 was 0.471, 0.498 and 0.825 for P1, P2 and P3, so the gain does not come from tuning on the dev questions. Answer quality improved as well, from 0.673 to 0.769 to 0.810 key-fact coverage (Table 5, Fig. 12).

Pipeline 3 fixed Q23: it found Article 18 and answered "10 years" with the right source. It did not fix everything:

- **Q11:** Article 99(6) was now a candidate (#7 after the first stage), but the cross-encoder moved it down to #8, so the answer still said "whichever is higher". The re-ranker does not seem to understand that "start-up" in the question makes the SME paragraph the important one.
- **Q46:** the exception list went from #2 after the first stage to #5 after re-ranking.
- **Q27:** the answer cited the right article but dropped the dataset-labelling exception.

7 Takeaways and limitations

Key findings

- Re-ranking fixed real cases, mostly role limits and exceptions hidden in lists. It also broke some (Q20, Q46), caused a wrong-role answer (Q23), and it can never recover a clause the first stage did not retrieve.
- On this document the main problems were the recitals and the recall of the first stage, not the choice of re-ranker. Fixing those moved MRR from 0.50 to 0.82, far more than a bigger model did.
- Cross-encoding 10 pairs costs about 17 ms, which is cheap next to a 5 s LLM call. Over the whole collection it would not scale, so two stages is the right design.

			top 3			MRR@10			MRR@10 re-ranked	
Group	Configuration	Chunks	hit 1st	hit re-rk	all re-rk	1st stage	re-rk	Recall pool	dev	test
chunking	reference (as in §5–6)	747	0.633	0.653	0.633	0.474	0.502	0.776	0.506	0.498
chunking	chunks ≤120 tokens	1610	0.592	0.694	0.673	0.483	0.558	0.796	0.577	0.539
chunking	chunks ≤400 tokens (truncated)	467	0.449	0.551	0.551	0.340	0.444	0.653	0.503	0.387
chunking	no chunk header	747	0.469	0.592	0.551	0.378	0.496	0.735	0.505	0.487
corpus	binding text only	458	0.776	0.796	0.776	0.651	0.736	0.837	0.748	0.725
pool	re-rank Top 20	747	0.633	0.694	0.673	0.474	0.534	0.837	0.542	0.527
pool	re-rank Top 30	747	0.633	0.694	0.673	0.474	0.534	0.837	0.541	0.527
pool	re-rank Top 50	747	0.633	0.714	0.694	0.474	0.552	0.898	0.581	0.525
first stage	bi-encoder bge-small	747	0.551	0.612	0.571	0.403	0.500	0.673	0.508	0.491
first stage	bi-encoder e5-base	747	0.592	0.694	0.653	0.452	0.564	0.816	0.567	0.561
first stage	bi-encoder bge-base	747	0.612	0.694	0.653	0.460	0.560	0.796	0.569	0.551
first stage	hybrid dense + BM25	747	0.694	0.735	0.714	0.512	0.563	0.857	0.585	0.543
re-ranker	cross-encoder MiniLM-L12	747	0.633	0.694	0.673	0.474	0.537	0.776	0.545	0.530
re-ranker	cross-encoder bge-reranker-base	747	0.633	0.755	0.735	0.474	0.574	0.776	0.616	0.533
re-ranker	cross-encoder bge-reranker-v2-m3	747	0.633	0.694	0.673	0.474	0.555	0.776	0.581	0.529
combined	binding + hybrid + Top 30	458	0.735	0.857	0.816	0.668	0.790	0.980	0.792	0.788
combined	binding + hybrid + Top 30 + L12	458	0.735	0.837	0.796	0.668	0.818	0.980	0.812	0.825
combined	binding + hybrid + Top 30 + bge-v2-m3	458	0.735	0.857	0.816	0.668	0.828	0.980	0.831	0.826
combined	binding + ≤120 tok + hybrid + Top 30 + L12	1038	0.735	0.837	0.796	0.672	0.756	0.939	0.765	0.748
combined	all text + ≤120 tok + hybrid + Top 30 + L12	1610	0.714	0.755	0.714	0.531	0.588	0.939	0.613	0.564

Figure 10. Every configuration I ran. "1st" = first-stage order; "re-rk" = after the cross-encoder; "recall pool" = share of questions whose gold clause was among the candidates the cross-encoder saw. Unless stated otherwise: MiniLM-L6 bi-encoder, ≤230-token chunks with headers, all text, top 10, MiniLM-L6 cross-encoder.

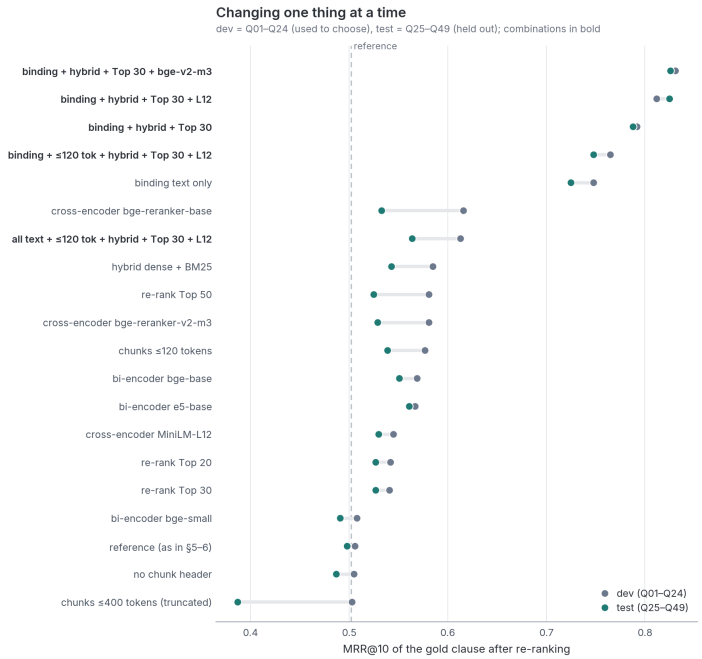


Figure 11. MRR@10 after re-ranking for every configuration, on dev and test questions. Dashed line: the assignment pipeline.

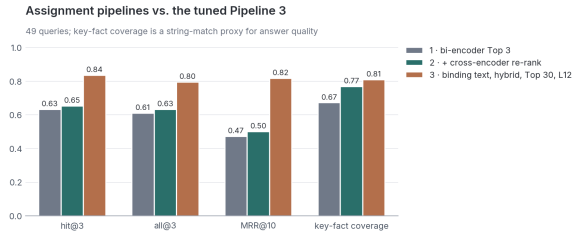


Figure 12. The three pipelines: retrieval metrics and key-fact coverage of the answers.

Limitations.

- The key-fact check is a rough proxy. Q02 scores 0 in every pipeline because no answer uses the exact words "untargeted scraping", even though all three answers are right.
- The gold labels only count binding text. Q19, for instance, was answered correctly ("2 February 2025") from Recital 179 but counts as a miss.
- I used 49 questions and one LLM.
- Ollama on a GPU is not fully deterministic: when I re-generated 4 answers, 2 came out with slightly different wording.

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A All 49 questions

Rank of the best gold chunk in each pipeline ("–" = not in the top 10). P1 = bi-encoder, P2 = re-ranked top 10, P3 = binding text + hybrid + top 30 + MiniLM-L12. The question type is in grey.

ID	Question	Gold	P1	P2	P3
Q01	Is it allowed to use AI to detect the emotions of employees at their workplace? (negation)	Art 5(1)	3	3	1
Q02	Is it legal to build a face recognition database by scraping photos from the internet? (negation/keyword-trap)	Art 5(1)	4	2	1
Q03	Can an AI system be used to predict whether a person will commit a crime? (negation)	Art 5(1)	1	2	1
Q04	When are police allowed to use real-time remote biometric identification in public spaces? (exception)	Art 5(1)	–	–	8
Q05	Can real-time biometric identification be used before a judge or authority has approved it? (exception)	Art 5(3)	4	4	2
Q06	Is every AI system listed in Annex III automatically considered high-risk? (exception)	Art 6(3)	–	–	1
Q07	Do providers of open-source general-purpose AI models have to prepare technical documentation? (exception)	Art 53(2)	5	5	9
Q08	Do deepfakes have to be labelled even when they are part of a satirical or artistic work? (exception)	Art 50(4)	3	2	1
Q09	Does the AI Act apply to AI systems used only for military purposes? (scope-negation)	Art 2(3)	3	3	1
Q10	Does a person using an AI system for a personal hobby have obligations under the AI Act? (scope)	Art 2(10)	–	–	4
Q11	What is the maximum fine for a start-up that puts a prohibited AI system on the market? (corner-case)	Art 99(3) + Art 99(6)	1	1	1
Q12	What is the fine for giving misleading information to the authorities? (penalty)	Art 99(5)	1	1	1
Q13	What fines can be imposed on providers of general-purpose AI models? (penalty/role)	Art 101(1)	1	1	1
Q14	What are the obligations of a deployer, as opposed to a provider, of a high-risk AI system? (role-limit)	Art 26	2	1	1
Q15	When is a distributor or deployer treated as the provider of a high-risk AI system? (role-limit)	Art 25(1)	1	1	1
Q16	What must an importer check before placing a high-risk AI system on the market? (role-limit)	Art 23(1)	4	1	1
Q17	Which deployers must carry out a fundamental rights impact assessment? (role-limit)	Art 27(1)	3	3	2
Q18	How much training compute makes a general-purpose AI model count as having systemic risk? (keyword-trap)	Art 51(2)	1	1	1
Q19	From what date do the bans on unacceptable AI practices start to apply? (keyword-trap)	Art 113	–	–	–
Q20	Do people have to be told that they are chatting with a bot? (synonym)	Art 50(1)	1	5	1
Q21	Is an AI system used for credit scoring considered high-risk? (exception)	Annex III pt 5	3	2	1
Q22	Does a person affected by a decision made with a high-risk AI system have a right to an explanation? (control)	Art 86(1)	1	2	1
Q23	For how long must providers keep the technical documentation of a high-risk AI system? (control)	Art 18(1)	–	–	1
Q24	What is the definition of an AI system? (control)	Art 3(1)	3	1	1
Q25	Are SMEs and start-ups given lower fines under the AI Act? (corner-case)	Art 99(6)	3	1	1
Q26	Is it prohibited to use AI that exploits the vulnerabilities of children or elderly people? (negation)	Art 5(1)	–	–	1
Q27	Can a company use biometric data to sort people by their race or sexual orientation? (negation)	Art 5(1)	5	2	1
Q28	Is social scoring of citizens banned? (negation)	Art 5(1)	3	2	1
Q29	How quickly must a provider report a serious incident that caused a person's death? (exception)	Art 73(4)	1	1	1
Q30	What is the deadline for reporting a serious incident? (control)	Art 73(2)	3	2	1
Q31	Does an employer have to inform workers before using a high-risk AI system in the workplace? (role-limit)	Art 26(7)	3	2	1
Q32	Can a high-risk AI system be used without completing the conformity assessment? (exception)	Art 46(1)	–	–	6
Q33	Do AI-generated news articles need to be disclosed if a human editor reviewed them? (exception)	Art 50(4)	1	2	1
Q34	Must people be informed when an emotion recognition system is used on them? (role-limit)	Art 50(3)	3	1	1
Q35	Is an AI tool that screens job applications and filters CVs high-risk? (control)	Annex III pt 4	1	1	1
Q36	Does the AI Act apply to AI being developed and tested before it is released? (scope)	Art 2(8)	–	–	2
Q37	What happens to high-risk AI systems already on the market before the rules apply? (corner-case)	Art 111(2)	1	1	1
Q38	Does the deepfake disclosure obligation apply when the content is used by law enforcement to investigate crimes? (exception)	Art 50(4)	2	1	1
Q39	Are there exceptions to the ban on emotion recognition at work? (exception)	Art 5(1)	5	4	1
Q40	Can a provider avoid the high-risk obligations by documenting that its Annex III system is not risky? (exception)	Art 6(3) + Art 6(4)	–	–	1

Q41	What fine does a deployer face for not complying with its obligations? (penalty/role)	Art 99(4)	1	1	1
Q42	Is AI used to assess students' exams or admission considered high-risk? (control)	Annex III pt 3	1	2	1
Q43	Can AI be used to decide on asylum applications? (control)	Annex III pt 7	1	3	1
Q44	Which uses of AI by law enforcement for lie detection are high-risk? (role-limit)	Annex III pt 6	2	4	2
Q45	Is facial recognition allowed for unlocking a phone or verifying identity? (exception)	Annex III pt 1	–	–	4
Q46	Can police use biometric identification to find a missing child? (exception)	Art 5(1)	2	9	5
Q47	Are AI systems used for detecting financial fraud classified as high-risk? (exception)	Annex III pt 5	5	2	1
Q48	How long do providers have to keep logs automatically generated by a high-risk AI system? (control)	Art 19(1)	1	1	1
Q49	When will most of the AI Act's rules start to apply? (keyword-trap)	Art 113	–	–	–